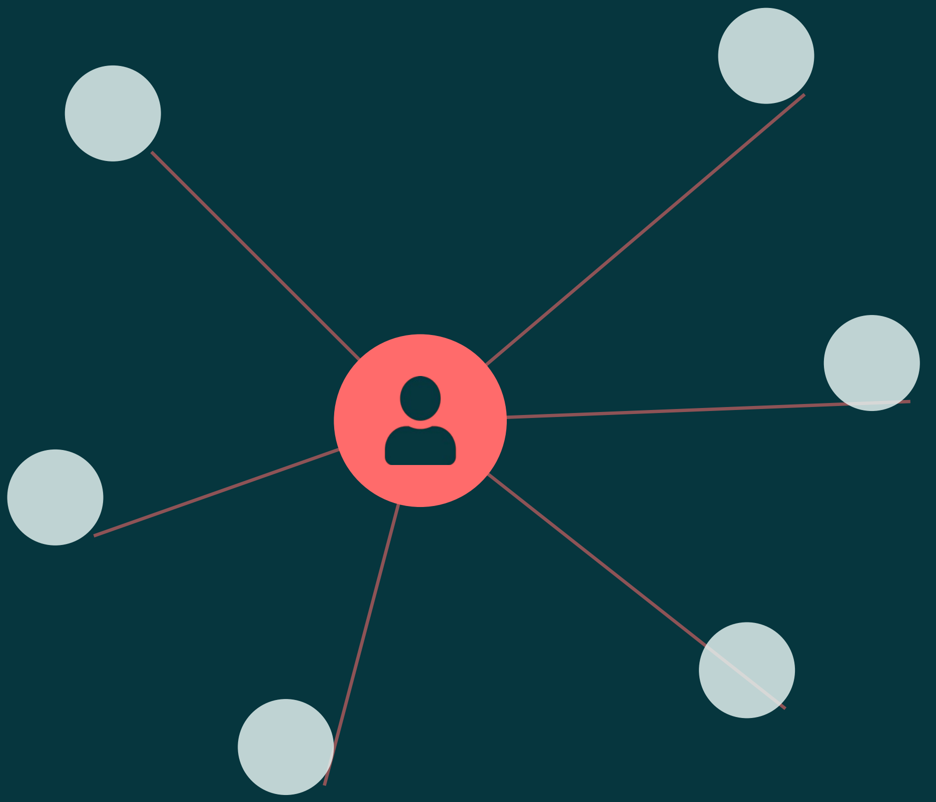
E-COMMERCE · A FRIENDLY INTRO

Product Recommendation Systems in E-Commerce

How They Actually Work — A Friendly Introduction

For Aspiring Data Scientists



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Helping shoppers find what they didn't know to search for

The discovery problem

Picture a store with a billion shelves. No one can browse it all.

Choice overload

Faced with endless options, people freeze and often buy nothing.

Why it matters

Good recommendations help people find things — and that drives sales and loyalty.

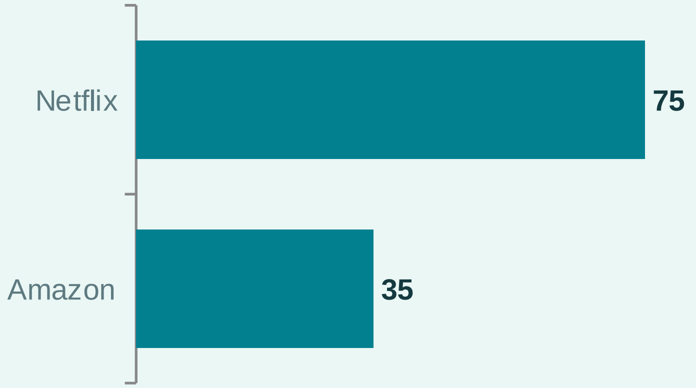
~35%

of Amazon sales tied to recommendations

~75%

of Netflix viewing comes from recommendations

Activity driven by recommendations



Widely cited estimates — useful as ballpark figures.

WHAT A RECOMMENDER DOES

Predict relevance, then rank

A recommender is a system that guesses what you'll like and shows you those items first. Think of a knowledgeable shop assistant who remembers everyone.

Input

Information about you, the items, and the moment (what you clicked, time, device).

→

Model

A formula that gives each item a relevance score for you.

→

Ranked output

The items sorted from most to least likely to interest you.

In short: data goes in, a score comes out, and the highest scores are shown first.

TYPES OF RECOMMENDATIONS

The same engine, many storefront surfaces

Personalized

“Recommended for you” — unique to each person.

Frequently bought together

The phone case that goes with the phone.

Trending & popular

What everyone's buying right now.

Similar items

“More like this one” you're looking at.

Recently viewed

A handy reminder of what you already saw.

CORE APPROACHES

How a recommender decides what to show

Four main ideas. Each answers “what should we show?” a different way.

Content-based

Liked a sci-fi novel? Here's another sci-fi novel. Matches by item features.

Collaborative filtering

People similar to you liked X, so you might too. Learns from behavior, not features.

Embeddings

Turn users and items into points in space; nearby points = good matches.

Hybrid

Combine the above to cover each other's weak spots.

Advanced systems use deep learning and session models that react to what you're doing right now.

THE DATA THAT POWERS IT

Two kinds of signal: what users say vs. what they do

Explicit signals

Users tell us directly

- Star ratings you give
- Reviews you write
- Likes & wishlists
- Preferences you select

Implicit signals

We observe behavior

- Clicks and taps
- Things you buy
- How long you look
- Items added to cart

Explicit = you tell us directly (rare but clear). Implicit = we observe behavior (plentiful but noisy).

THE COLD-START PROBLEM

When there's no history to learn from

New user

A brand-new user: we have no history, so we can't personalize yet.

New item

A brand-new item: nobody has interacted with it, so we can't learn from behavior.

Common workarounds

- Show popular items to everyone at first
- Ask a quick “what are you into?” at sign-up
- Recommend new items by their features (content-based)
- Use clues like location, device, and time of day

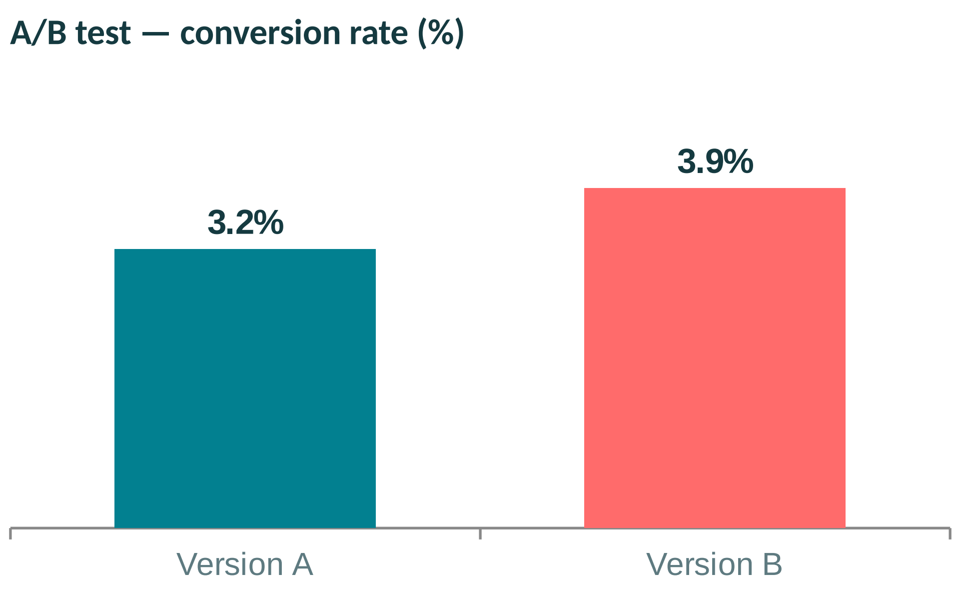
Test offline to shortlist, test online to decide

Offline metrics (lab tests)

- Precision@k, Recall, NDCG
- Did good items land near the top of the list?

Online metrics (real users)

- Click-through, conversion, revenue
- Measured with A/B tests on live traffic



Offline tests narrow the options; A/B tests on real users make the final call.

A REAL-WORLD ARCHITECTURE

Retrieve broadly, rank precisely, then apply the rules

STEP 1

Candidate generation

Quickly shrink millions of items to a few hundred maybes.

Millions → hundreds

STEP 2

Ranking

Carefully score those few hundred for this exact user.

Hundreds → dozens

STEP 3

Re-ranking & rules

Add variety, remove out-of-stock, apply promos.

Dozens → top few

Slow, heavy work is done ahead of time (batch). The final ranking happens live, in milliseconds (real-time).

CHALLENGES & ETHICS

Relevance is not the only thing that matters

Filter bubbles

Only seeing similar things can trap users in a loop.

Popularity bias

Popular items get recommended more, getting even more popular.

Diversity vs. relevance

Spot-on but repetitive, or varied but risky? Find a balance.

Privacy

Personalization uses data — handle it responsibly and with consent.

Fairness

Avoid systematically favoring some users or sellers.

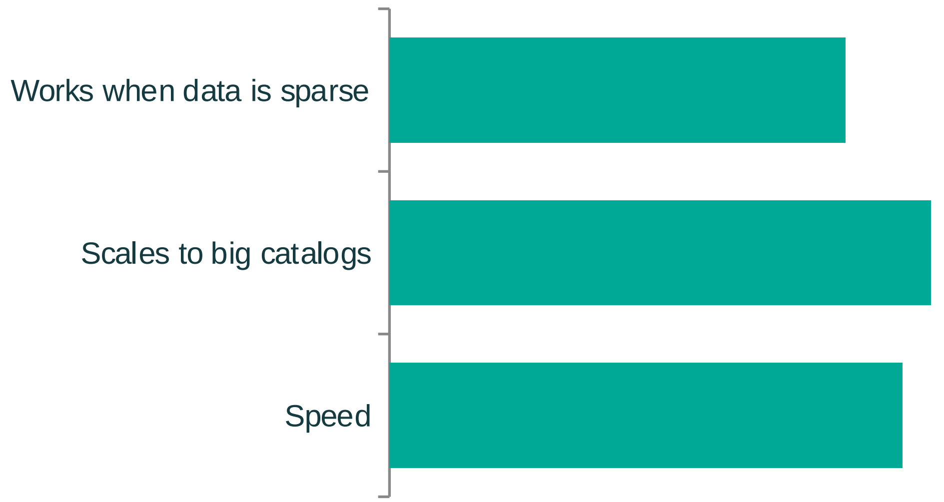
CASE STUDY

Amazon's item-to-item collaborative filtering

The approach

1. Instead of comparing users, compare items: which products co-occur?
2. Precompute an “items like this” list for every product, offline
3. At serve time, just look it up — fast, even with huge catalogs

Why item-based won (illustrative)



A famous, beginner-friendly example: a clever shift in perspective made recommendations fast and scalable.

KEY TAKEAWAYS

What to carry out of this session

1. Recommenders predict what you'll like and rank items for you
2. Two big ideas: content-based (features) and collaborative (behavior)
3. Implicit data is everywhere; explicit data is clearer but rarer
4. Watch out for cold-start — new users and new items
5. Always measure: offline metrics first, A/B tests to decide

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Now you have the map. Next stop: build your first recommender!

Thank you